الأكاديمية السعودية الرقمية
SAUDI DIGITAL ACADEMY

WeCloudData

PROJECT REPORT



JobDataPipeline

SUBMITTED BY : TEAM C

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1. WHAT WE BUILT

A repeatable, automated data pipeline that collects technology job postings in Saudi Arabia from five sources, cleans and standardizes them, and publishes an analysis-ready star schema in Snowflake that feeds a Power BI dashboard.

Each scheduled run loads only new files and cleans only new rows, so the dataset stays current without reprocessing history. The raw data is kept unchanged in a data lake, so every number in the dashboard can be traced back to its source.

5	1,331	707	723	15	69/69
sources	job postings	companies	skills	locations	checks passed

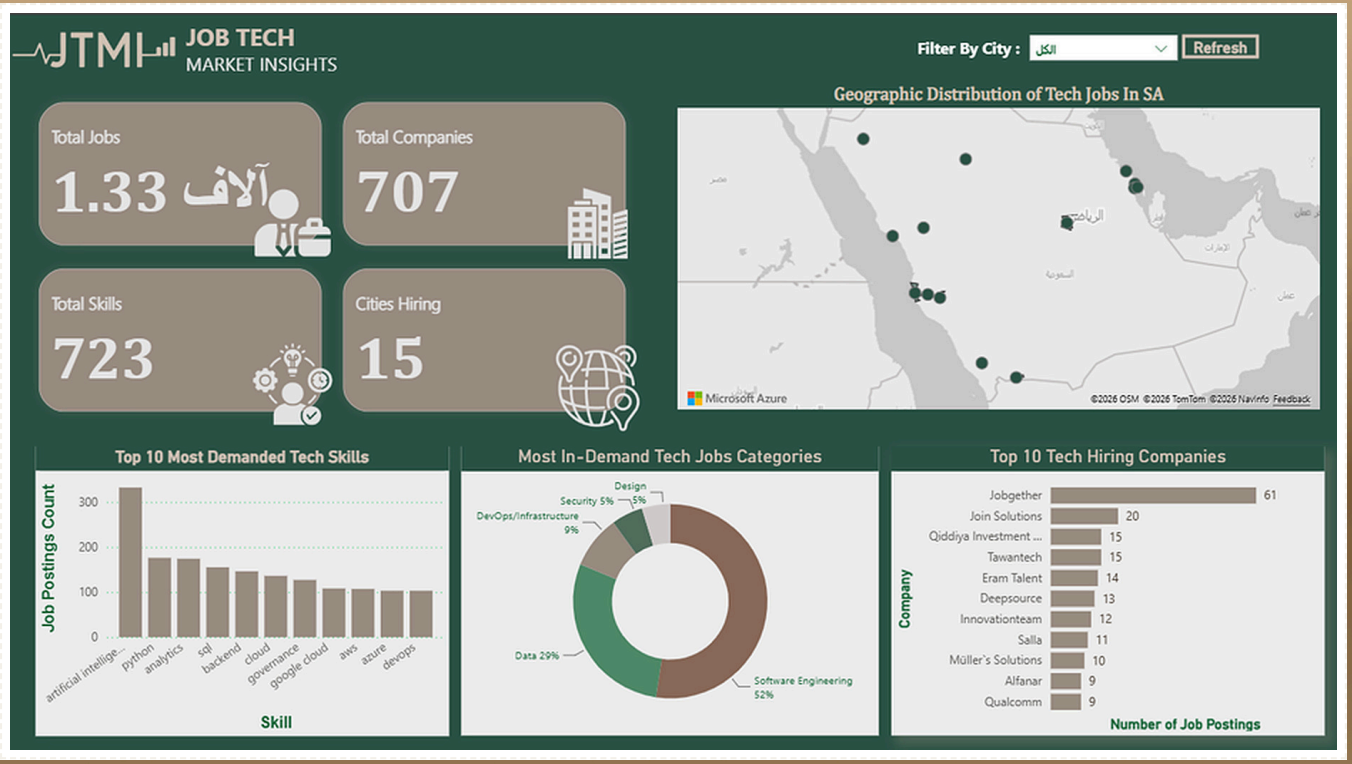


FIGURE 1. THE POWER BI DASHBOARD BUILT ON THE FINAL DATASET.

2. WHO IT IS FOR

Stakeholder	What they get
Job seekers and tech talent	The most demanded skills, trending job categories, and the top hiring companies, to guide their career.
Recruiters and HR teams	Benchmarks for role requirements, current market demand, and the top hiring companies.
Labor market analysts	The geographic distribution of tech jobs across Saudi Arabia, and market trends and patterns.

3. HOW IT WORKS

The pipeline follows a Medallion Architecture: data moves through three layers, Bronze (raw), Silver (staging), and Gold (marts), and no later stage modifies an earlier layer in place.

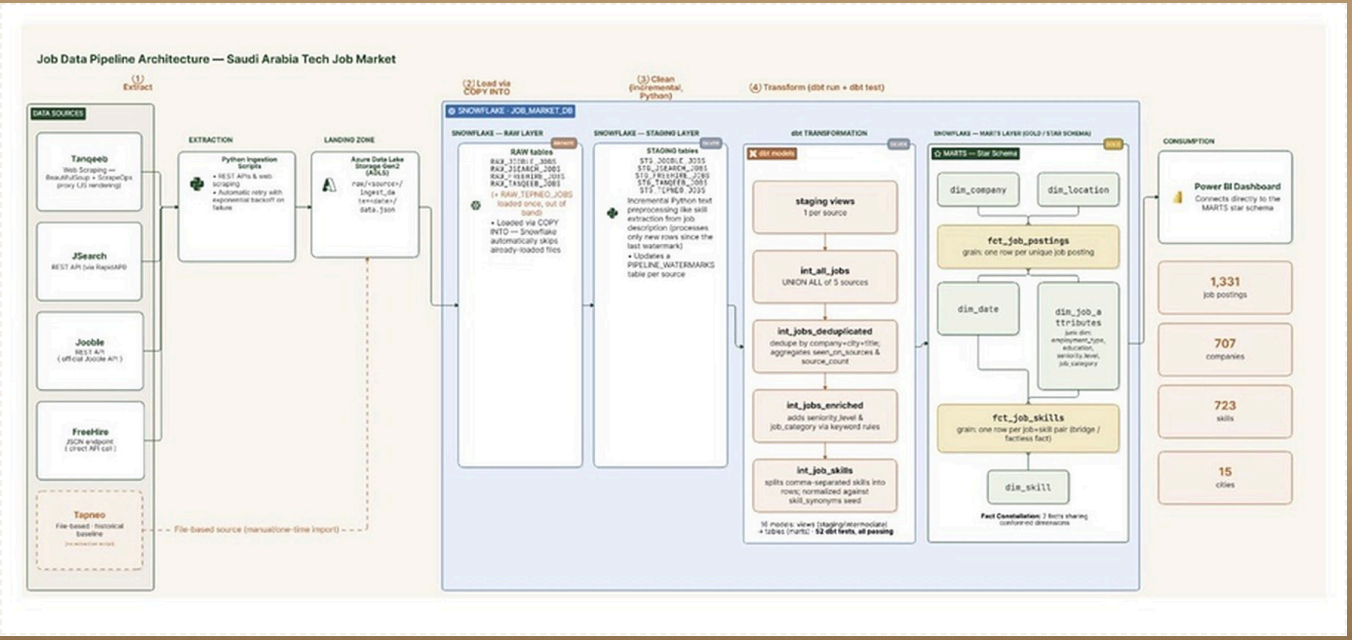


FIGURE 2. END-TO-END ARCHITECTURE, FROM THE FIVE SOURCES TO THE DASHBOARD.

Layer	Where	What it holds
Bronze	ADLS **raw/<source>/ingest_date=<date>/** and Snowflake **RAW** tables	Raw JSON as extracted, one table per source
Silver	Snowflake **STAGING** tables, then dbt staging and intermediate models	Cleaned records in one unified schema; merged, deduplicated, and enriched
Gold	Snowflake **marts**	Star schema with 2 fact tables and 5 dimensions

3.1 EXTRACT

Each of the four live sources has a dedicated Python extractor. It stops at a fixed target per run and waits 0.5–2 seconds between requests to respect rate limits. Tapneo is a one-time historical file loaded directly into the raw layer.

Source	Access method	Per-run target
Jooble	Official REST API (POST with API key), retries with exponential backoff	50 jobs
JSearch	RapidAPI (GET), skips a keyword when the quota is reached	100 jobs
FreeHire	JSON endpoint with offset pagination, retries on server errors	150 jobs
Tanqeeb	Web scraping with BeautifulSoup and a ScrapeOps proxy	20 jobs
Tapneo	Static historical file, loaded once	—

3. HOW IT WORKS

3.2 LOAD INTO SNOWFLAKE

New files are loaded from the data lake into one RAW table per source with **COPY INTO**. Snowflake tracks previously loaded files, so re-running the load does not reprocess or duplicate them.

```
COPY INTO RAW_JOOBLE_JOBS (RAW_DATA, SOURCE_FILE, INGEST_DATE)
FROM (SELECT $1, METADATA$FILENAME, CURRENT_DATE() FROM @adls_stage/jooble/)
```

3.3 INITIAL CLEANING WITH PYTHON

Python handles the heavy text work. It reads only rows newer than each source's watermark, maps all sources into one unified schema, and then updates the watermark, so this stage is fully incremental.

- **Tech scope:** removes non-tech roles, such as civil or architectural engineering.
- **Text:** removes HTML, symbols, and repeated headers such as "Job Description".
- **Standardization:** unifies city names, employment types, and education levels.
- **Skills:** extracts skills with a dictionary of 300+ terms and regular expressions.
- **Language:** applies a language check to Tanqeeb's Arabic listings.

3.4 ORCHESTRATION WITH PREFECT

One Prefect flow, **job-data-pipeline**, runs nine tasks in three sequential stages. **wait_for** makes each stage wait for the one before it, and every task retries once after 30 seconds.

1 · Extract
4 tasks, one per live source

→

2 · Load
1 task, **COPY INTO** for 4 RAW tables

→

3 · Clean
4 tasks, one per source

The flow pulls the code from GitHub at runtime with an authenticated access token, so it runs the same way on any machine. It is scheduled every two days at 11:00 Riyadh time. dbt is run as a separate step after the flow completes.

3.5 TRANSFORMATION WITH DBT

Layer	What it does
Staging 5 views	One **stg_*** model per source standardizes column names and formats.
Intermediate 4 views	**int_all_jobs** combines the five sources with UNION ALL. **int_jobs_deduplicated** merges the same job across sources on company, city, country, location, and title. **int_jobs_enriched** adds seniority level and job category. **int_job_skills** splits skills into rows and unifies synonyms with the **skill_synonyms** seed.
Marts 7 tables	The final star schema: 5 dimensions and 2 fact tables.

3. HOW IT WORKS

3.6 DATA MODEL

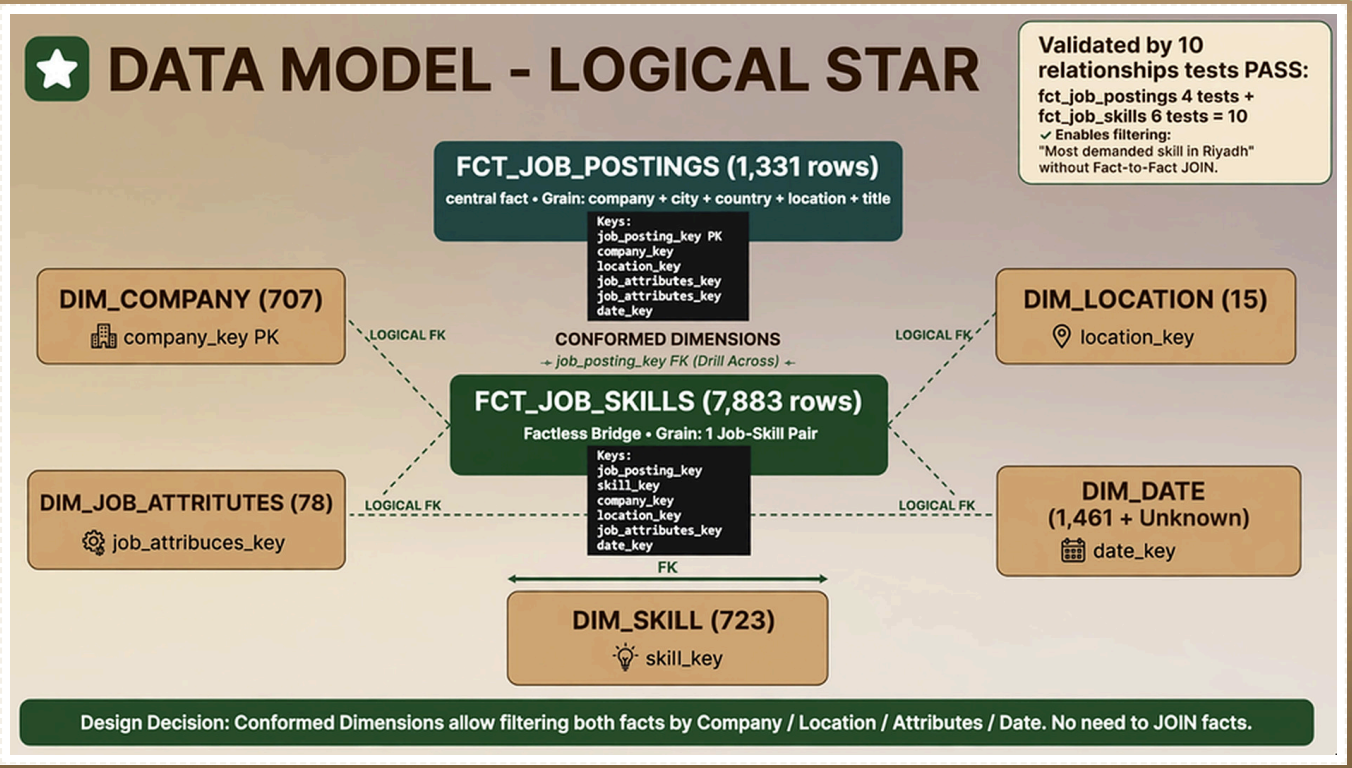


FIGURE 3. STAR SCHEMA: TWO FACT TABLES SHARING THE SAME DIMENSIONS.

- **Two fact tables:** **fct_job_postings** has one row per job (1,331), and **fct_job_skills** has one row per job and skill (7,883). One table would repeat each job for every skill and inflate job counts. Sharing the same dimensions makes this a Fact Constellation.
- **Junk dimension:** **dim_job_attributes** combines employment type, education, seniority, and category, instead of four one-column tables.
- **Unknown Member pattern:** missing values become "Unknown Company" or "Not Specified", so no record is lost.



FIGURE 4. DBT LINEAGE, FROM THE FIVE STAGING TABLES TO THE FINAL FACT AND DIMENSION TABLES.

3. HOW IT WORKS

3.7 DATA QUALITY

Automated dbt tests run with every build, so a model is complete only after its data passes validation.

Test	Count	What it checks
not_null	33	Job titles, company names, and every key are filled
unique	7	Every dimension key and **job_posting_key** is unique
relationships	10	Every foreign key in both fact tables points to a real row
accepted_values	1	Employment type values are valid (warning only)
Composite uniqueness	1	No repeated job in the deduplicated model

The latest full build ran 1 seed, 9 views, 7 tables, and 52 tests in 34.69 seconds: **69 of 69 checks passed, with 0 warnings and 0 errors.** Jobs grew from 1,256 to 1,331 between two refreshes, and every test still passed.

```
09:00:03  69 of 69 START test relationships_fct_job_skills_skill_key__skill_key__ref_dim_skill_  [RUN]
09:00:04  69 of 69 PASS relationships_fct_job_skills_skill_key__skill_key__ref_dim_skill_  [PASS in 0.23s]
09:00:04
09:00:04  Finished running 1 seed, 7 table models, 52 data tests, 9 view models in 0 hours 0 minutes and 34
.69 seconds (34.69s).
09:00:04
09:00:04  Completed successfully
09:00:04
09:00:04  Done. PASS=69 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=69
```

FIGURE 5. OUTPUT OF THE LATEST DBT BUILD.

4. AI COMPONENTS

The current pipeline does not use AI models. Tech-job filtering and skill extraction use keyword dictionaries and regular expressions, which keeps every result explainable and repeatable. The final dataset is ready to serve as a foundation for AI-powered applications, such as natural-language job search, job recommendations, and career insights.

5. WHAT WORKED WELL

- **Exploring sources early:** testing each source first showed its limits, so each one got a suitable method: official APIs, a JSON endpoint, and scraping only for Tanqeeb, which has no public API.
- **Running from GitHub:** after Prefect could not access our private repository, a GitHub access token and pulling code at runtime made every run repeatable.
- **Incremental loading:** **COPY INTO** file tracking and the watermark mean each run processes only new data.
- **Tests that caught real defects:** orphaned company and date keys, and a mismatch between the posting key and the deduplication grain, were fixed before reaching the dashboard.

6. WHAT DID NOT WORK OR IS STILL INCOMPLETE

- **Partial deduplication:** Jooble's text snippet changes on every crawl, so we deduplicate on job title, which is only a partial fix. About 7 duplicate URLs remain because of small title differences.
- **Refresh frequency:** the flow runs every two days, so it does not yet meet a 24-hour freshness target.
- **Separate dbt step:** dbt runs outside the Prefect flow, so the marts are not refreshed automatically after each run.
- **Limited historical tracking:** changes to a posting over time are not kept.
- **Coverage:** the scope is Saudi Arabia only.
- **Housekeeping:** the **skill_synonyms** seed loads into the PUBLIC schema instead of **staging_dbt**, and dbt-utils shows non-blocking deprecation warnings.

7. WHAT WE WOULD IMPROVE NEXT

Improvement	Result
Run **dbt build** as the last Prefect task, and schedule the flow daily	Fully automated refresh within 24 hours
Add sources for the UAE and other GCC countries	New tech jobs across the region reach the dashboard
Refine deduplication with URL-based and fuzzy matching	Removes the remaining near-duplicates
Add dbt snapshots (SCD Type 2)	A full history of changes to each posting
Update dbt-utils syntax and move the seed to **staging_dbt**	No warnings and a consistent schema layout
Build AI-powered applications on the marts	Job search, recommendations, and career insights

SUMMARY

The pipeline turns fragmented job postings from five sources into one curated, tested dataset of 1,331 unique jobs, 707 companies, 723 skills, and 7,883 job-skill links, already powering an interactive Power BI dashboard.